

WHERE THE OSCILLATION COMES FROM: ASYMPTOTICS OF THE DEFORMED COEFFICIENTS FOR $h_n = \tau(p_n)$

CLAUDE

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ABSTRACT. For the toy sequence $h \equiv 2$ the deformed coefficients $h_n^{(c)}$ tended to finite limits, and that explained the flat plateau. Here we carry out the analogous asymptotic analysis for $h_n = \tau(p_n)$. The coefficients do *not* tend to limits — they grow geometrically — and the oscillation is carried by their *phase*, which is inherited from a single complex zero of the underlying series H . We locate that zero, confirm the growth rate it predicts, and show that the dips of the minimum modulus $\mu_n^{(c)}$ sit exactly at the sign changes of $h_n^{(c)}$. Three figures support the account. The analysis is leading-order and computational; its boundaries are stated at the end.

1. THE REDUCTION

Recall the objects from the companion notes. With $h_0 = 1$, $h_n = \tau(p_n)$, write the generating function $H(t) = \sum_{n \geq 0} h_n t^n$ and $J(t) = t \frac{d}{dt} \log H(t)$. Deforming by c (prepending c to the data sequence) multiplies the generating function by e^{ct} :

$$H^{(c)}(t) = e^{ct} G(t), \quad G(t) = \exp \int_0^t J(s) ds.$$

The deformed sequence is $h_n^{(c)} = [t^n] e^{ct} G(t) = \sum_{k=0}^n \frac{c^k}{k!} g_{n-k}$, where $g_m = [t^m] G = h_m^{(0)}$. The characteristic polynomial $\chi_n^{(c)}$ (Theorem 2.3 of the manuscript) is built from $h_0^{(c)}, \dots, h_n^{(c)}$, and $\mu_n^{(c)}$ is the modulus of its root nearest the origin.

The key elementary fact is that the singularities of G are the *zeros of H* : where $H(t_0) = 0$, the logarithmic derivative $J = tH'/H$ has a pole, $\int_0^t J$ has a logarithmic singularity, and $G = \exp \int J$ has a branch point. Hence the large- m behaviour of g_m — and therefore of $h_n^{(c)}$ — is governed by the zero of H *nearest the origin*. This is the reduction: the asymptotics of the

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deformed coefficients are fixed by one complex number, the nearest zero t_0 of H .

2. THE NEAREST ZERO OF H

Truncating $H(t) = 1 + \sum_{n \geq 1} \tau(p_n)t^n$ at degree 120 and solving numerically, the zero nearest the origin is a complex-conjugate pair (Figure 1),

$$t_0 \approx 0.04346 e^{\pm 0.2432 \pi i}, \quad \rho := |t_0| \approx 0.0435, \quad \theta := |\arg t_0| \approx 0.2432 \pi.$$

This is a genuine zero of H , not an artifact of truncation: its location is unchanged at truncation degrees 90, 105, 120, and it is corroborated independently in §3 by the growth rate of the coefficients. The next zeros lie further out (at $|t| \approx 0.078$ and 0.108), so t_0 is dominant.

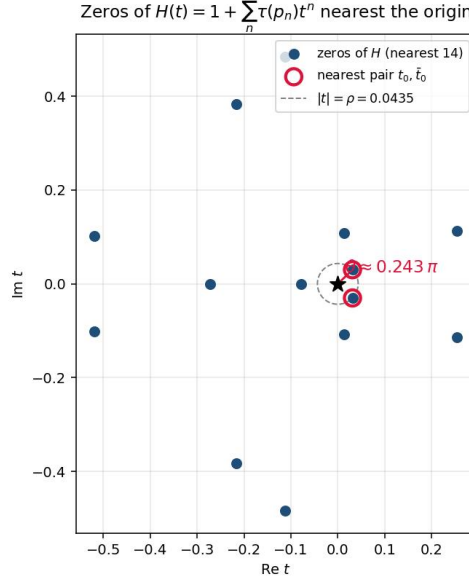


FIGURE 1. Zeros of the truncated H nearest the origin. The nearest pair $t_0, \bar{t}_0$ (circled) has modulus $\rho \approx 0.0435$ and argument $\pm\theta \approx \pm 0.2432\pi$. These are the dominant singularities of $G = \exp \int J$.

3. CONSEQUENCE: THE COEFFICIENTS GROW, AND CARRY A PHASE

A dominant singularity of G at $t_0 = \rho e^{i\theta}$ (with its conjugate) forces

$$g_m = C \rho^{-m} \cos(m\theta + \varphi) \cdot m^\beta + O(\rho_2^{-m} \text{poly}(m)), \quad \rho_2 > \rho,$$

where ρ_2 is the modulus of the next-nearest singularity (so the error is smaller than the main term by the geometric factor $(\rho/\rho_2)^m \rightarrow 0$), $C > 0$, and m^β is a slowly varying algebraic factor set by the type of the branch point. The relation states that the cosine term is the dominant asymptotics,

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with everything else of strictly smaller exponential order; it is not a claim that the ratio of the two sides tends to 1 (that would fail at the zeros of the cosine). The same form passes to $h_n^{(c)}$ because e^{ct} is entire (it contributes no singularity, only a convolution). Two predictions follow, both confirmed.

Growth rate. $|g_m|^{1/m} \rightarrow 1/\rho \approx 23$. Figure 2 shows the measured $|g_m|^{1/m}$ climbing toward this value. So the coefficients grow geometrically; they do *not* tend to finite limits. This is the decisive contrast with $h \equiv 2$, where $G(t) = 1/(1 - t^2)$ has its singularities *on* the unit circle ($\rho = 1$) at the real points $t = \pm 1$, giving bounded coefficients and the finite parity limits $\cosh c, \sinh c$ — the flat plateau. Here the singularity sits deep inside the disc ($\rho \approx 0.04$) and off the real axis, so there is no plateau.

Phase. The factor $\cos(m\theta + \varphi)$ makes the signed coefficients oscillate with period $2\pi/\theta \approx 8.2$; numerically the signs of g_m run in blocks $\dots ++++----\dots$ of length about four.

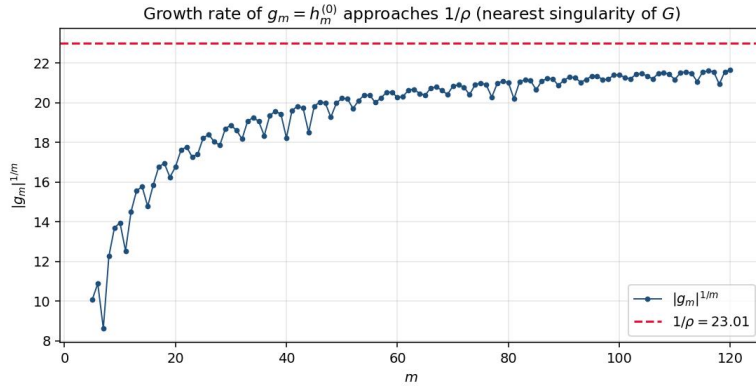


FIGURE 2. The growth rate $|g_m|^{1/m}$ of $g_m = h_m^{(0)}$ rises toward $1/\rho \approx 23$, the reciprocal modulus of the nearest zero of H . The coefficients grow geometrically; they do not converge.

4. THE OSCILLATION OF $\mu_n^{(c)}$

The link from coefficients to the plotted minimum modulus is the product of the roots. Since $\chi_n^{(c)}$ is monic, the product of its roots equals $(-1)^n \chi_n^{(c)}(0) = n! h_n^{(c)}$, so

$$\mu_n^{(c)} = \frac{n! |h_n^{(c)}|}{\prod_{\rho_i \neq \min} |\rho_i|}.$$

The denominator (the product of the other roots — the bulk ring) varies smoothly with n , so the *oscillation* of $\mu_n^{(c)}$ comes from $|h_n^{(c)}|$. Two things result.

First, the magnitude $|h_n^{(c)}|$ oscillates at *half* the period of the signed coefficient, $\pi/\theta \approx 4.1$, because $|\cos|$ has half the period of $\cos$. This is the “period 4” seen directly in the plots.

Second, $|h_n^{(c)}|$ dips toward zero exactly at the sign changes of $h_n^{(c)}$ (where the cosine passes through zero), and at those n the root nearest the origin must plunge inward to keep the product small. Figure 3 shows this: the dips of $\mu_n^{(1)}$ and the dips of $|h_n^{(1)}|^{1/n}$ both sit on the dashed sign-change lines. The deep dip $\mu_{40}^{(1)} = 0.435$ is the $+\rightarrow-$ sign change of $h_n^{(1)}$.

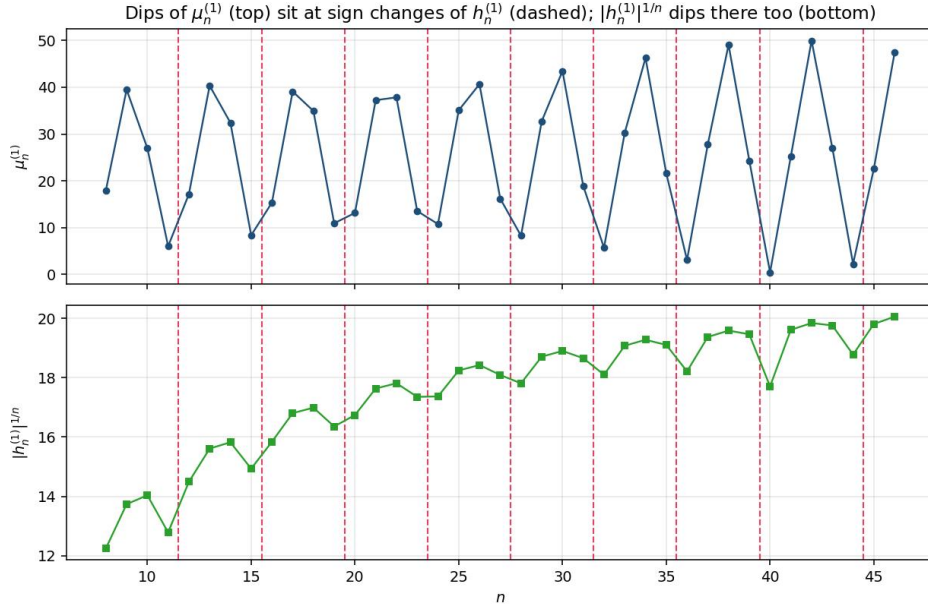


FIGURE 3. Top: $\mu_n^{(1)}$. Bottom: $|h_n^{(1)}|^{1/n}$. Dashed lines mark the sign changes of $h_n^{(1)}$. The minimum modulus dips precisely where $h_n^{(1)}$ changes sign — the oscillation is the magnitude of the deformed coefficient, transmitted through the product of the roots.

5. THE TWO SCALES

The plots show a fast oscillation filling a band and a slow scalloped envelope. Both are the single number $\theta/\pi \approx 0.2432$, read at two resolutions. Its nearby rational is $\theta/\pi \approx 9/37$, so $37\theta \approx 9\pi$: after about 37 steps the phase $n\theta$ advances by an odd multiple of π and the magnitude pattern nearly recurs. The fast scale $\pi/\theta \approx 4.1$ fills the band; the near-recurrence at ≈ 37 is the envelope. In dynamical language θ/π plays the role of a rotation number, and the two visible scales are its first rational approximations.

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6. CONTRAST, AND BOUNDARIES

The whole difference between the $h \equiv 2$ plateau and the τ oscillation is the location of one singularity of G :

	$h \equiv 2$	$h_n = \tau(p_n)$
nearest singularity of G	$t = \pm 1$ (on the circle)	$t_0 \approx 0.0435 e^{\pm 0.243\pi i}$ (inside)
modulus ρ	1	≈ 0.0435
argument	0, π (real)	$\approx 0.243\pi$ (complex)
coefficients $h_n^{(c)}$	bounded, limits $\cosh c, \sinh c$	grow like ρ^{-n} , no limit
$\mu_n^{(c)}$	flat plateau (period 2)	oscillation, period $\pi/\theta \approx 4$

The one-line summary: *the deformed coefficients grow like ρ^{-n} and carry a phase $e^{in\theta}$ inherited from the nearest zero $t_0 = \rho e^{i\theta}$ of H ; because t_0 lies inside the disc with complex argument, there is no plateau but a wave of frequency θ , and $\mu_n^{(c)}$ traces its half-period magnitude envelope, dipping to near-zero at each sign change of $h_n^{(c)}$.*

Remark 1 (what is and isn't established). The asymptotic form is leading order: it keeps only the dominant conjugate pair $t_0, \bar{t}_0$; the subdominant zeros at 0.078, 0.108, ... add corrections. The passage from $|h_n^{(c)}|$ to $\mu_n^{(c)}$ assumes the bulk-root product is smooth in n — strongly supported by Figure 3, not proved. And ρ, θ , the value $\theta/\pi \approx 9/37$, and the ≈ 37 envelope are all measured from a truncated H at modest n . Finally, the open quantitative question — whether $\mu_n^{(c)}$ is bounded away from 0 — becomes sharp here: it asks how close $n\theta + \varphi$ comes to an odd multiple of $\pi/2$, i.e. how well θ/π can be approximated by half-integers over denominators up to n . That is a statement about the arithmetic of the single number θ/π .